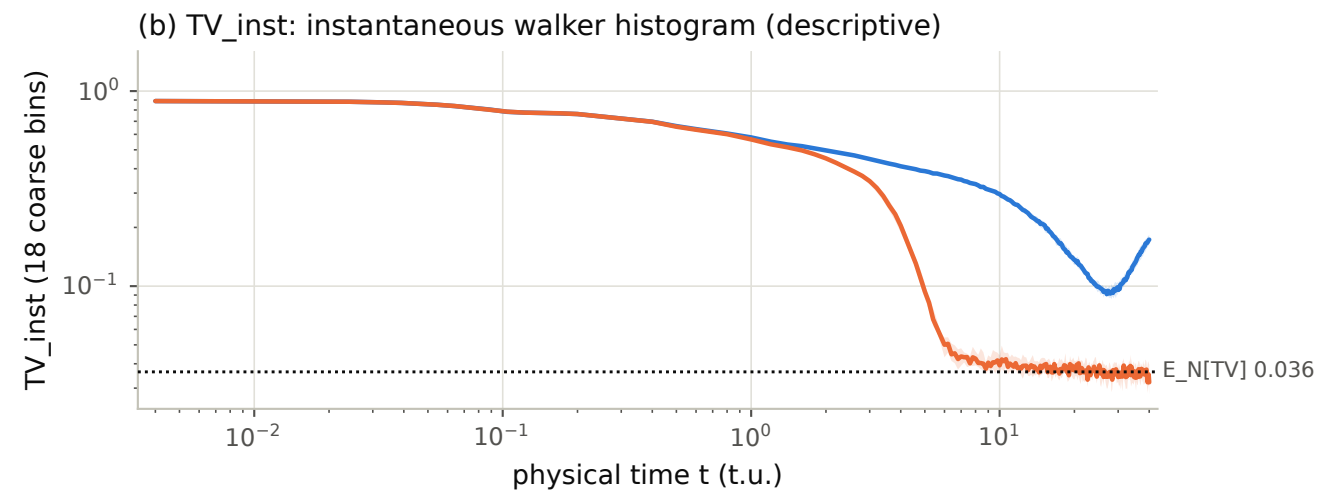
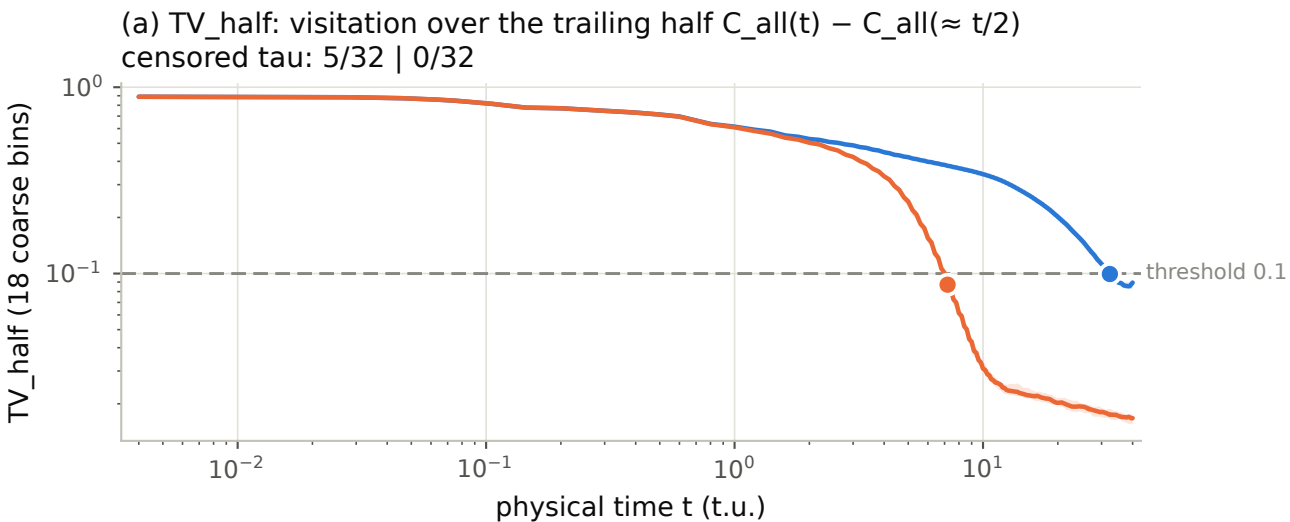


# Gateway family matched\_free\_energy/shift · N = 2048 · T\_N = 40 t.u.

n\_steps = 1,600,000 · h = 2.5e-05 · B = N n\_steps = 3.277e+09 walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Marginal convergence: TV\_half and TV\_inst with the finite-N floor vs physical time



- ABF (n = 32)
- ABF+FR (n = 32)
- interquartile range
- TV\_half threshold 0.1 (marker: persistent crossing of the median curve)
- finite-N floor  $E_N[TV] = 0.0363$  (exact, multinomial(N, uniform))

Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC\_PLAN section-4 quantities, unsmoothed.

TV\_half uses the saved step nearest  $t/2$  (or the run start) as the window's lower edge (eqb\_metrics.tv\_half); tau for the marginal is defined on TV\_half. TV\_inst is never used for tau: at small N it is dominated by discreteness (17/18 at  $N = 1$ ).

eqb\_plot\_config/2 · config shift.json · 2026-10-10 14:20 UTC